

Cogniland

A Partially-Observed Navigation Testbed for the
Mechanistic Study of *Belief* in World Models and Recurrent Agents

Project overview

August 20, 2026

Abstract

Cogniland is a research codebase built around one idea: give a reinforcement-learning agent a task where it *cannot see everything at once*, so that to act well it must build and hold an internal *belief* about the hidden state of the world; then open the agent up and study that belief directly. This document is a self-contained tour of the project. We describe the current flagship environment—the **bridge_tunnel fork.wall** task, a small grid world in which a hidden “map category” must be inferred from terrain seen early, remembered across a blind corridor, and finally read out as a left-or-right door choice. We then describe the two kinds of agents we train on it (a recurrent policy and a *world model*), the interpretability tools we use to read and edit their beliefs (linear probes, activation steering, and manifold geometry), and the experiments and results those tools produce. Throughout we favour plain language and give the math needed to reproduce every claim. The recurring lesson is that the agent’s belief is real, causal, and geometrically structured—but also that getting a world model to learn the task at all exposed a subtle reward-design pitfall that we document and fix.

Contents

1	Introduction: why a “belief” testbed	2
2	The environment: bridge_tunnel / fork.wall	2
2.1	The task in one paragraph	2
2.2	The hidden variable: map category	3
2.3	Observation: an egocentric window	4
2.4	Actions and dynamics	4
2.5	The memory corridor (task design)	4
2.6	Reward, and a reward-shaping pitfall	4
2.7	A fixed map dataset	5
3	The agents	5
3.1	Recurrent PPO with a GRU memory	5
3.2	World models: DreamerV3 / R2-Dreamer	6
3.3	What we mean by “belief”	6
4	Methods: reading and editing belief	7
4.1	Probing: decoding the belief distribution	7
4.2	Steering: editing the belief to test causality	7
4.3	Geometry: is belief a direction or a manifold?	7
4.4	Dreams: decoding the imagined future	8

5 Experiments and results	8
5.1 Training: the task is solvable, and requires memory	8
5.2 Experiment 1: the belief <i>causes</i> the fork decision	9
5.3 Experiment 2: the belief shapes the dream	10
5.4 Experiment 3: belief is a <i>curved</i> manifold, but a robust one	11
5.5 Reading the whole picture	13
6 Discussion and lessons	14
7 Conclusion	14

1 Introduction: why a “belief” testbed

Most interesting decision problems are *partially observed*: the thing you need to know is not written on the current observation. A driver at a fork does not see the destination; a rat in a maze does not see the reward arm; a language model predicting the next token does not see the author’s intent. Formally these are *partially observable Markov decision processes* (POMDPs) [Kaelbling et al.(1998)]. The classical result for POMDPs is that the optimal action depends not on the current observation o_t but on the *belief state*—a probability distribution over the hidden state given the entire history $o_{1:t}, a_{1:t-1}$. A good agent must therefore compute something belief-like and carry it forward in memory.

Recurrent neural networks do exactly this in practice: their hidden state is a learned, compressed summary of history, and there is now strong evidence—in neuroscience [Mante et al.(2013), Wong & Wang(2006), Sussillo & Barak(2013)], in meta-RL [Mikulik et al.(2020), Lambrechts et al.(2022)], and in large models [Li et al.(2023), Shai et al.(2024)]—that these hidden states encode recoverable, sometimes even Bayes-optimal, beliefs about latent variables. **Cogniland exists to make this phenomenon concrete and manipulable.** We want a task where (i) the hidden variable is simple and known to us (so we can check what the agent believes against the truth), (ii) the belief must genuinely be *held in memory* (so it is not a trivial reflex), and (iii) the belief must be *used* to drive a discrete decision (so it is behaviourally load-bearing). The `fork.wall` environment (§2) is engineered to satisfy all three.

Once such a task exists, we can ask sharper questions than “does the agent succeed?”. We can ask: *where* is the belief stored? Is it *causal* (does editing it change behaviour)? Does it lie on a simple geometric object? And in a *world model*—an agent that learns to imagine the future—does the belief shape what the agent dreams? Sections 4 and 5 develop the tools and answers.

2 The environment: `bridge_tunnel` / `fork.wall`

2.1 The task in one paragraph

The agent is a single point that starts at the left edge of a rectangular grid and must reach a goal door on the right. Early in its journey it walks past patches of **terrain**—water and rock—whose *amounts* secretly encode the map’s *category*. It then crosses a stretch of empty grass (the *memory corridor*), where no terrain is visible, and arrives at a **wall** with a narrow passage. Beyond the wall are two doors, one near the top and one near the bottom. Exactly which door counts as success depends on the category the agent saw earlier. Because the terrain has scrolled out of view by

the time the agent reaches the doors, it must have *remembered* the category to choose correctly. Figure 1 shows one map per category.

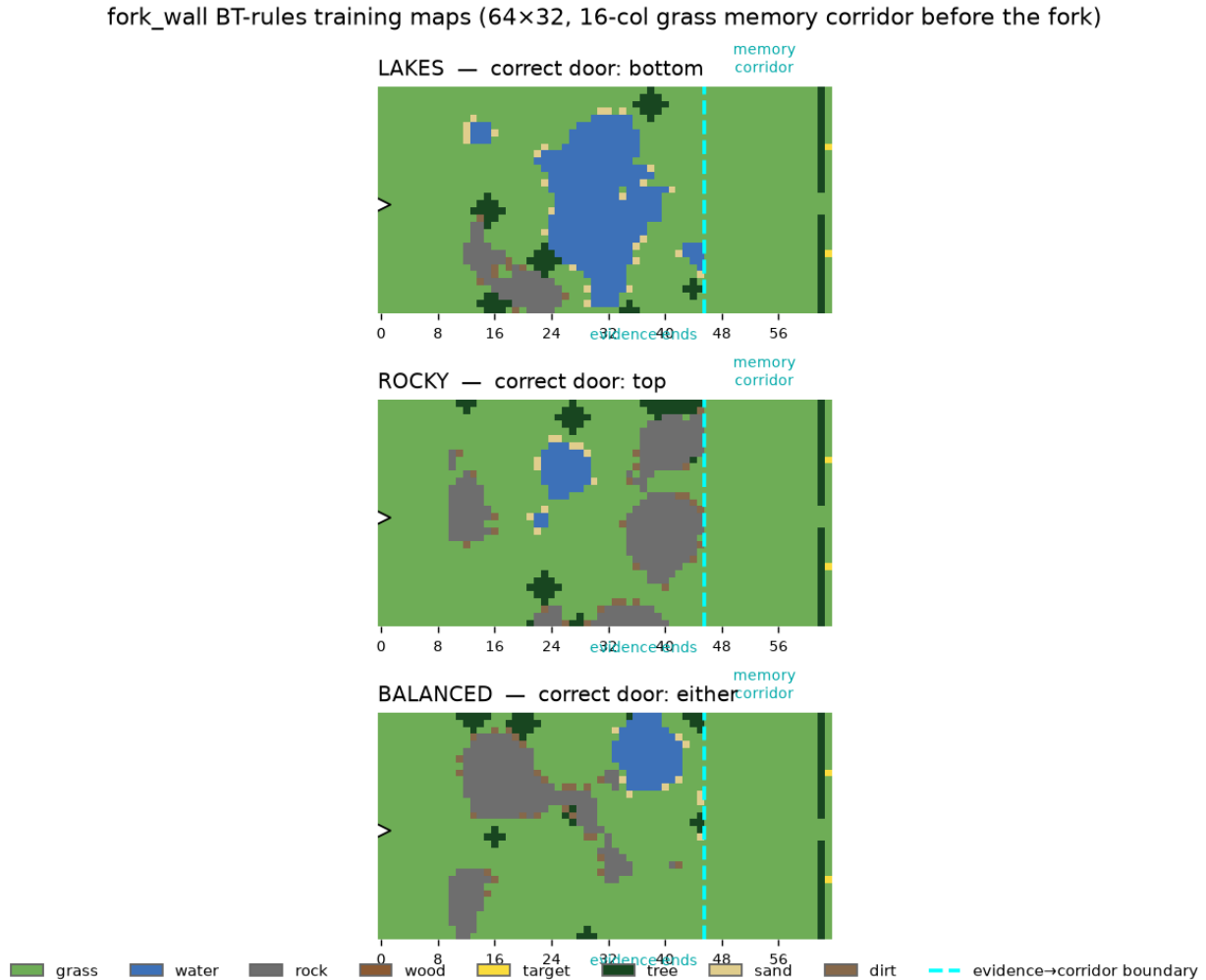


Figure 1: Three maps from the fixed training set, one per category. Spawn is the white marker on the left. Category evidence (water for **lakes**, rock for **rocky**, both for **balanced**) is confined to the left of the cyan line; a grass *memory corridor* then runs to the fork wall and its top/bottom doors (yellow). By the fork the evidence has left the agent’s egocentric view, so the door choice requires memory rather than a glance.

2.2 The hidden variable: map category

Every map belongs to one of three categories, defined by how much water versus rock its terrain contains:

- **lakes**: water-dominant. The *correct* door is the **bottom**.
- **rocky**: rock-dominant. The correct door is the **top**.
- **balanced**: roughly equal water and rock. *Either* door counts.

The category is never handed to the agent as an input; it is only implicit in the terrain. It is the “hidden state” of our POMDP, and “what the agent believes the category is” is the belief we study throughout.

2.3 Observation: an egocentric window

The agent never sees the whole map. At each step it receives a square, agent-centred crop of the terrain of side $V = 21$ cells, together with a few scalar signals (which way the goal lies, how far into the episode we are). Write the terrain as $T \in \{0, \dots, K-1\}^{H \times W}$ with $K = 9$ tile types (grass, water, rock, wood, target, tree, sand, dirt, and an out-of-bounds pad), $H = 32$, $W = 64$. The observation is

$$o_t = \left(\underbrace{\text{onehot}\left(T\left[r_t - \frac{V-1}{2} : r_t + \frac{V-1}{2}, c_t - \frac{V-1}{2} : c_t + \frac{V-1}{2}\right]\right)}_{V \times V \times K \text{ one-hot crop}}, \underbrace{s_t}_{\in \mathbb{R}^5} \right), \quad (1)$$

where (r_t, c_t) is the agent’s cell and cells outside the map are filled with the out-of-bounds tile. Flattened, this is a vector of dimension $V^2K + 5 = 21^2 \cdot 9 + 5 = 3974$. Crucially, because the window has radius $\lfloor V/2 \rfloor = 10$, terrain more than 10 cells behind the agent is *invisible*. This single fact is what makes the memory corridor (§2.5) do its job.

2.4 Actions and dynamics

The action space is $\mathcal{A} = \{\text{up, down, left, right, build, mine}\}$ (six discrete actions). Moves are grid steps; **build** turns a faced water cell into a walkable bridge and **mine** turns a faced rock cell into grass. In the variant used here (“BT-rules”, i.e. commitment disabled) the tools are always available and the agent can also simply walk around obstacles, so the decision that matters is purely *which door* it goes to. An episode ends when the agent steps onto either door, or after a step budget of 800.

2.5 The memory corridor (task design)

An earlier version of this map placed terrain right up to the fork, so the agent could see the evidence and the doors *in the same window* and pick the door by reflex—no memory needed. We fix this with a **memory corridor**: a band of $g = 16$ pure-grass columns immediately before the wall, with all category terrain confined to its left. Let c_{wall} be the wall column and c_{ev} the rightmost terrain column; the construction guarantees $c_{\text{wall}} - c_{\text{ev}} \geq g$. Since the view radius is $10 < g$, when the agent stands at the fork ($c_t \approx c_{\text{wall}} - 1$) the nearest evidence at c_{ev} is $\geq g - 1 > 10$ cells away and therefore outside the window. Concretely the evidence leaves view once $c_t - 10 > c_{\text{ev}}$, giving the agent on the order of $c_{\text{wall}} - (c_{\text{ev}} + 10) \approx 6$ “blind” timesteps in which the correct action depends only on memory. This turns a reflex into a genuine working-memory problem.

2.6 Reward, and a reward-shaping pitfall

The reward at each step is a sum of three parts:

$$r_t = \underbrace{-\lambda_{\text{slack}}}_{\text{time cost}} + \underbrace{\rho [\text{stepped onto the correct door}]}_{\text{goal bonus}} + \underbrace{\kappa (\Phi(s_t) - \gamma_{\text{sh}} \Phi(s_{t+1}))}_{\text{shaping}}, \quad (2)$$

with $\lambda_{\text{slack}} = 0.01$, $\rho = 3.0$, $\kappa = 0.015$, and the *potential* $\Phi(s) = -\text{ctg}(s)$, where $\text{ctg}(s)$ is the number of steps from the agent’s cell to the correct door (a cost-to-go computed by breadth-first search on the terrain). Shaping of the form $F = \gamma_{\text{sh}} \Phi(s') - \Phi(s)$ is the classic *potential-based reward shaping* of Ng, Harada, and Russell [Ng et al.(1999)], which provably does not change the optimal policy when γ_{sh} equals the RL discount γ .

The pitfall. Potential-based shaping is policy-invariant in theory but not *exploration*-invariant in practice. At a fixed state ($s_{t+1} = s_t$, i.e. the agent stands still), the shaping term is

$$\kappa(\Phi - \gamma_{\text{sh}}\Phi) = \kappa(1 - \gamma_{\text{sh}})(-\Phi) = \kappa(1 - \gamma_{\text{sh}})\text{ctg}(s) > 0. \quad (3)$$

Near the spawn, ctg is large (≈ 70), so with $\gamma_{\text{sh}} = \gamma = 0.99$ this is $0.015 \cdot 0.01 \cdot 70 \approx 0.0105$ per step, which *exceeds* the time cost $\lambda_{\text{slack}} = 0.01$. **The agent is literally paid to do nothing.** We observed exactly this failure: a DreamerV3 world model, trained from scratch, converged to a policy that never left the spawn tile (maximum column reached = 0) while collecting positive reward and timing out every episode. On-policy PPO happened to explore its way out of this basin, but the world model, which learns its policy *in imagination*, sat in it comfortably; raising the exploration bonus did not help.

The fix. We set $\gamma_{\text{sh}} = 1$, turning the shaping into *pure progress*:

$$\kappa(\Phi(s_t) - \Phi(s_{t+1})) = \kappa(\text{ctg}(s_t) - \text{ctg}(s_{t+1})). \quad (4)$$

Now standing still earns exactly zero shaping (net reward $-\lambda_{\text{slack}} < 0$), a step toward the goal earns $+\kappa$, and a step away earns $-\kappa$; there is no stationary way to accumulate reward, so the only route to positive return is to reach a door. Empirically this immediately removed the basin: the world model began leaving the spawn and reaching doors, and evaluation success rose from a flat 0 toward the same regime PPO reaches. We flag this because it is a general and easily-missed lesson: *discounted potential shaping can create a “loitering” optimum whenever the potential is large and the step cost is small*, and an imagination-based learner is especially prone to fall into it.

2.7 A fixed map dataset

So that every agent trains and is tested on identical data, we pre-generate a fixed dataset of 6000 maps—2000 per category—and split it 80/20 into 4800 *train* and 1200 *test* maps. Maps are procedurally generated from simplex-noise heightmaps thresholded to hit the target water/rock fractions, then post-processed to carve the memory corridor and guarantee the correct door is reachable. Training draws maps uniformly from the train split; all reported success numbers are on the disjoint test split, so nothing evaluated was ever seen in training.

3 The agents

We study two agents on identical data. They differ in *what* they learn: the first learns only to act; the second additionally learns to *imagine*.

3.1 Recurrent PPO with a GRU memory

The first agent is a recurrent policy trained with Proximal Policy Optimization (PPO) [Schulman et al.(2017)]. Because the task is partially observed, the policy needs memory; we give it a Gated Recurrent Unit (GRU) [Cho et al.(2014)]. An encoder maps the observation o_t to a feature e_t ; the GRU updates a hidden state

$$h_t = \text{GRU}(e_t, h_{t-1}) \in \mathbb{R}^{128}, \quad (5)$$

and linear heads read the action distribution $\pi_\theta(a_t | h_t)$ and value $V_\theta(h_t)$ off h_t . The GRU hidden state h_t is the agent’s working memory and hence its belief; it is what a probe (§4.1) reads. PPO

optimizes the clipped surrogate

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(\varrho_t(\theta) \hat{A}_t, \text{clip}(\varrho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad \varrho_t(\theta) = \frac{\pi_\theta(a_t | h_t)}{\pi_{\theta_{\text{old}}}(a_t | h_t)}, \quad (6)$$

with advantages $\hat{A}_t$ from generalized advantage estimation. We optionally add a small *auxiliary head* that predicts the map category from h_t ; this does not change behaviour but makes the belief more linearly readable, and its accuracy (`belief_acc`) is a convenient training-time diagnostic.

3.2 World models: DreamerV3 / R2-Dreamer

The second agent is a *world model* in the DreamerV3 family [Hafner et al.(2023), Ha & Schmidhuber(2018)]. A world model learns a compact latent model of the environment’s dynamics and then trains its policy by “dreaming” —rolling that latent model forward and doing RL on the imagined trajectories, never touching the real environment during policy improvement. Concretely we use the in-repo `purejaxwm` port and an efficient PyTorch reimplementation, R2-Dreamer [Morihiro et al.(2026)]. The core is a *Recurrent State Space Model* (RSSM), whose state has a deterministic part d_t and a stochastic part z_t :

$$d_t = f_\phi(d_{t-1}, z_{t-1}, a_{t-1}) \quad (\text{deterministic recurrence, a GRU}) \quad (7)$$

$$\hat{z}_t \sim p_\phi(\hat{z}_t | d_t) \quad (\text{prior: what the model expects, no observation}) \quad (8)$$

$$z_t \sim q_\phi(z_t | d_t, e_t) \quad (\text{posterior: corrected by the encoded observation } e_t) \quad (9)$$

The world model is trained to (a) predict the next stochastic state (a KL between posterior and prior), (b) predict reward and episode continuation, and, in the reconstruction variant we use here, (c) *decode* the observation back from the latent state, $\hat{o}_t = \text{dec}_\phi(z_t, d_t)$. Writing the full latent (“feature”) as $b_t = [z_t; d_t] \in \mathbb{R}^{3840}$, the world-model loss is a variational bound of the form

$$\mathcal{L}^{\text{WM}} = \mathbb{E} \left[\underbrace{-\log \text{dec}_\phi(o_t | b_t)}_{\text{reconstruction}} + \underbrace{-\log p_\phi(r_t | b_t) - \log p_\phi(\text{cont}_t | b_t)}_{\text{reward/continue}} + \beta \underbrace{\text{KL}[q_\phi(z_t) \| p_\phi(\hat{z}_t)]}_{\text{dynamics}} \right]. \quad (10)$$

The actor and critic are trained purely on latent rollouts of the *prior* (imagination): starting from a real b_t , one repeatedly samples $a \sim \pi$, advances $d, \hat{z}$ with f_ϕ and p_ϕ , and optimizes the policy against imagined reward. This is what makes a world model special for us: its belief b_t is not only read by the policy—it *generates* the future the policy plans against. We can literally decode what the agent imagines.

Reconstruction vs. decoder-free. R2-Dreamer’s default objective is decoder-free (it replaces reconstruction with a redundancy-reduction term). For this project we deliberately train with the *reconstruction* objective (“`rep_loss=dreamer`”) so that the world model is genuinely generative and its dreams decode to legible maps—at the mild cost of extra compute. When we analyse an older decoder-free checkpoint, we instead fit a small decoder post hoc by frozen-backbone probing (it reaches 99.6% per-cell tile accuracy), which recovers the same picture.

3.3 What we mean by “belief”

Two notions of belief appear, and it helps to keep them straight.

- The **internal belief** is the recurrent state itself—the GRU hidden h_t for PPO, the RSSM feature $b_t = [z_t; d_t]$ for the world model. It is a high-dimensional vector, the physical substrate of memory.

- The **belief distribution** is a probability vector over the three map categories, $\mathbf{p}_t = (p_t^{\text{balanced}}, p_t^{\text{lakes}}, p_t^{\text{rocky}})$ Δ^2 . This is the “Bayesian” belief a reader expects: as evidence accumulates it should sharpen from near-uniform (uncertain) toward a one-hot (committed). We obtain it by *probing* the internal belief (§4.1).

The whole project is, in a sense, about the map from the first to the second, and about whether that map is causal and geometric.

4 Methods: reading and editing belief

4.1 Probing: decoding the belief distribution

A *probe* is a simple classifier fit to predict the hidden category from the internal belief [Ravichander et al.(2021), Li et al.(2023)]. Given a dataset of (belief, true-category) pairs $\{(b^{(i)}, c^{(i)})\}$ collected from rollouts, we fit two linear probes:

1. **Nearest class mean.** Compute the per-category centroid $\mu_c = \frac{1}{n_c} \sum_{i:c^{(i)}=c} b^{(i)}$ and predict $\hat{c}(b) = \arg \min_c \|b - \mu_c\|_2$. This is the simplest possible linear readout; its decision boundary is the perpendicular bisector between centroids.
2. **Logistic regression.** Fit a multinomial logistic model $\mathbf{p}(b) = \text{softmax}(Wb + w_0)$ with an ℓ_2 penalty chosen by cross-validation. This is a *discriminative* linear probe: it learns the directions that best separate the classes, and its softmax output is exactly the belief distribution $\mathbf{p}_t$.

The probe is deliberately linear: if a linear map recovers the category, the belief is stored in a linearly-accessible way, which is the property that makes the later steering experiments meaningful. Because the probe is trained on one set of episodes and evaluated on held-out ones, its accuracy measures genuine *decodability*, not memorization.

4.2 Steering: editing the belief to test causality

Decoding shows the belief is *present*; steering tests whether it is *causal*. The idea, following activation-editing work in language models [Turner et al.(2023), Zou et al.(2023), Meng et al.(2022)] and the causal-abstraction programme [Geiger et al.(2021)], is to *patch* the belief toward another category and see whether behaviour follows. We form a steering direction $\Delta_{s \rightarrow t}$ from source category s to target t and apply

$$b' = b + \alpha \Delta_{s \rightarrow t}, \quad (11)$$

then feed b' to the (frozen) actor and record its decision. Two choices of direction:

- **Difference of means:** $\Delta_{s \rightarrow t} = \mu_t - \mu_s$, the vector between class centroids.
- **Probe weights:** $\Delta_{s \rightarrow t} = w_t - w_s$, the difference of the logistic probe’s per-class weight vectors—the directions the classifier actually found most discriminative.

A clean causal result looks like: patching a belief from **rocky** toward **lakes** flips the agent’s door choice from top to bottom. Reporting both directions lets us separate “the belief is decodable” (a property of the probe) from “the belief is steerable” (a property of the representation)—these are not the same, as §5.2 shows.

4.3 Geometry: is belief a direction or a manifold?

Difference-of-means steering assumes the belief varies along a straight *line* between categories. But recurrent belief states are known to trace *curved, low-dimensional structures*—line attractors and rings—in their activation space [Sussillo & Barak(2013), Shai et al.(2024)]. If the belief lives on

a curved *manifold*, moving in a straight line between two beliefs leaves the manifold, landing on states the network never actually occupies. We test this directly, adapting the manifold-steering method of the Goodfire `causalab` framework [Geiger et al.(2021)].

Concretely we parameterize belief by a scalar *evidence coordinate* $\zeta \in [-1, 1]$ (signed by which door the trajectory heads to, scaled by progress), bin it, and take the mean belief in each bin as a control point $\{(\zeta_j, \bar{b}_j)\}$. We fit a natural cubic smoothing spline $\gamma : [-1, 1] \rightarrow \mathbb{R}^{3840}$ through these points (the 1-D analogue of a thin-plate spline, minimizing curvature $\int \|\gamma''\|^2$) [Reinsch(1967)], giving a smooth 1-D curve $\mathcal{M} = \{\gamma(u)\}$ in belief space. Two ways to steer from belief $\gamma(u_a)$ to $\gamma(u_b)$:

$$\text{linear: } b(t) = (1 - t) \gamma(u_a) + t \gamma(u_b), \quad (12)$$

$$\text{manifold: } b(t) = \gamma((1 - t)u_a + t u_b), \quad (13)$$

i.e. interpolate in *ambient space* (a chord) versus in the *intrinsic coordinate* (staying on the curve). We measure how far each path strays from the manifold via Gauss–Newton projection: for a point x , find $u^* = \arg \min_u \|\gamma(u) - x\|^2$ and report the residual $\|x - \gamma(u^*)\|$. If the belief manifold is curved, the linear chord will bow off it while the manifold path stays on it by construction.

4.4 Dreams: decoding the imagined future

Because the world model is generative, we can ask what it *imagines*. From a belief b_t we roll the prior forward under the actor for H steps and decode each latent to a map crop:

$$\hat{o}_{t+k} = \text{dec}_\phi(b_{t+k}), \quad b_{t+k+1} = [\hat{z}_{t+k+1}; f_\phi(d_{t+k}, \hat{z}_{t+k}, a_{t+k})], \quad a_{t+k} \sim \pi. \quad (14)$$

This yields a sequence of imagined observations—a “dream”—that we can look at directly and measure (e.g. what fraction of imagined tiles are water vs. rock, or where the goal door appears). Dreams let us ask the sharpest version of our question: does the belief not only label the map but *generate* a future consistent with that label?

5 Experiments and results

We first establish that the task is solvable and that belief is the mechanism, then progressively test how structured and how causal that belief is. Unless noted, analyses are run on a trained `fork_wall` BT-rules agent and report held-out numbers. The figures in this section are produced by the belief-analysis pipeline on a trained `fork_wall` BT-rules agent (held-out episodes); the memory-corridor world model of §2.6 is retrained on the fixed dataset and its figures will be regenerated identically once it saturates.

5.1 Training: the task is solvable, and requires memory

On the fixed dataset with the memory corridor, recurrent PPO reaches 100% **success** on held-out maps, with an auxiliary category-probe accuracy of ≈ 0.90 and efficient ~ 100 -step episodes—it walks the corridor, holds the category across the blind stretch, and picks the correct door. That the corridor makes this a genuine memory problem is visible in training: on the old evidence-at-the-fork maps the policy hit 100% almost immediately, whereas on the memory maps it climbs gradually as the recurrent state learns to carry the belief. The world model, once the reward-shaping pitfall of §2.6 is fixed, likewise leaves the spawn and its success rises from 0 into the same regime. The key methodological takeaways are (i) the environment is well-posed and solvable, and

(ii) an imagination-based learner is uniquely sensitive to farmable shaping and must be trained with the pure-progress form.

5.2 Experiment 1: the belief *causes* the fork decision

We collect beliefs at the moment just before the agent commits up or down, fit the probes of §4.1, and test steering.

The belief is decodable. The map category is linearly readable from the pre-decision belief well above the 33% chance level: the nearest-mean probe reaches $\approx 60\%$ and the logistic probe $\approx 67\%$. (These are deliberately weak linear readouts; a small nonlinear probe would score higher.)

Behaviour follows the *belief*, not the map. The decisive test is a within-model consistency check. On held-out episodes we compare the door the agent actually reached against (a) the true category and (b) the probe’s *decoded* category. The door matched the *decoded* belief $\approx 100\%$ of the time—slightly *tighter* than it matched the ground truth ($\approx 92\%$). In other words, on the episodes where the model’s belief was wrong, its *decision* was wrong in exactly the same way. The agent acts on what it believes, not on what is true—the signature of a genuine internal belief driving behaviour.

Decodable is not the same as steerable. Patching the pre-decision belief with the difference-of-means direction changes the action $\approx 92\%$ of the time but only lands on the *target* door $\approx 12\%$ of the time; using the logistic probe’s own discriminative direction raises the on-target rate to $\approx 38\%$ (a $3\times$ improvement). Two lessons: the category direction found by a discriminative probe is a much better steering handle than the naive centroid difference; and even so, the belief is not perfectly separable by linear arithmetic—category information is entangled with the “where am I in the episode” information the recurrent state also carries.

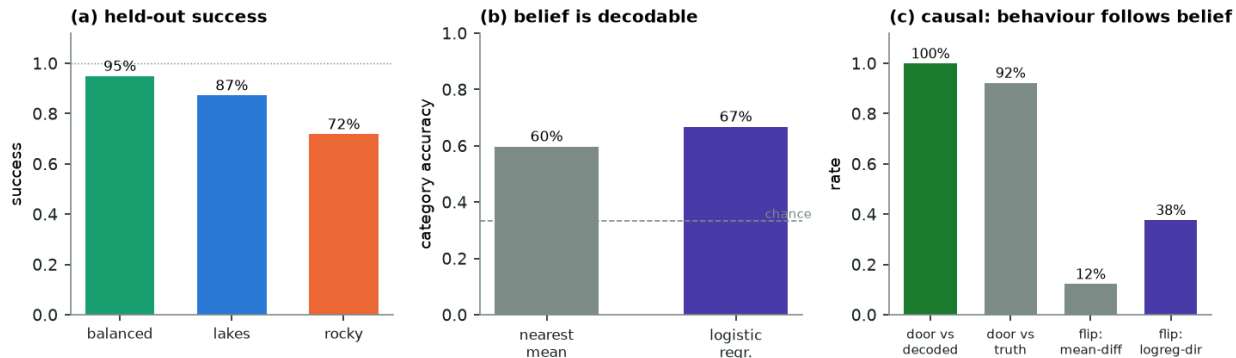


Figure 2: Experiment 1, quantitative. (a) Held-out success per category for the analysed agent. (b) The map category is linearly decodable from the pre-decision belief, well above the 33% chance line; a discriminative (logistic) probe beats the nearest-mean readout. (c) Behaviour follows the belief: the door reached matches the probe’s *decoded* category ($\approx 100\%$) even more tightly than the true category ($\approx 92\%$). Steering by the logistic probe’s own direction lands on the target door about $3\times$ more often than the naive centroid difference—but neither is perfect, confirming decodable $\neq$ steerable.

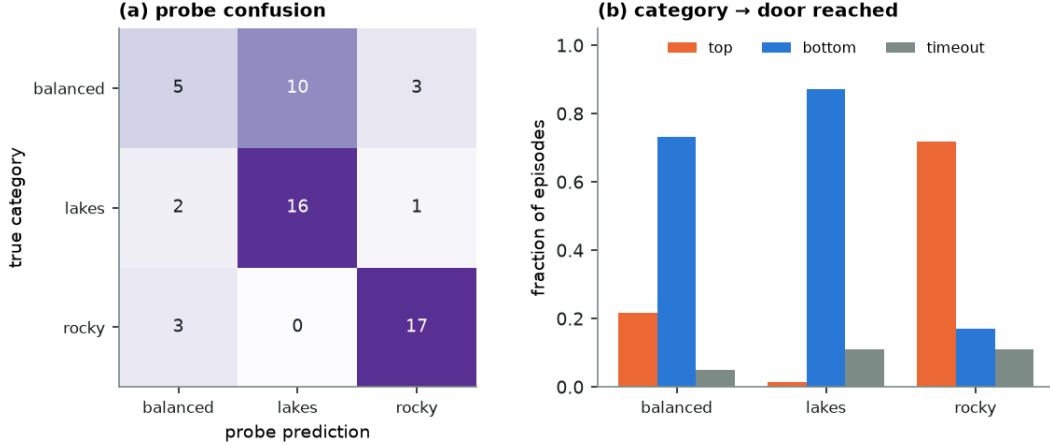


Figure 3: Experiment 1, structure. **(a)** Confusion matrix of the logistic probe (counts): the diagonal dominates, with the residual confusion concentrated on the ambiguous **balanced** class. **(b)** The category→door behaviour the belief drives: **rocky** goes to the top door, **lakes** to the bottom, **balanced** to either; timeouts are rare.

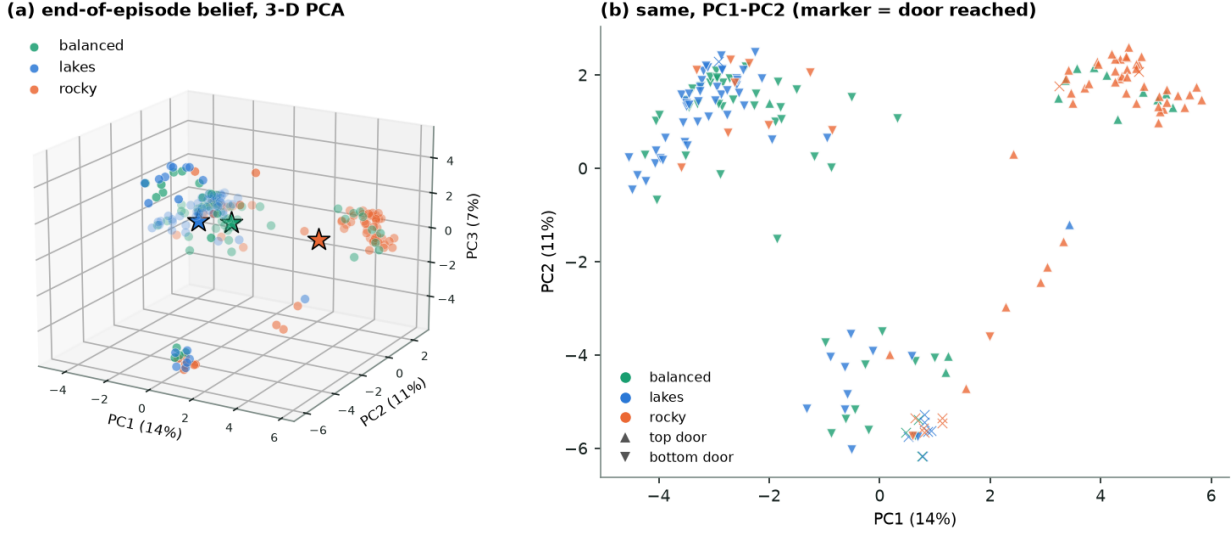


Figure 4: The belief state *at the end of the episode* (the pre-decision state at the fork), projected onto its first three principal components. **(a)** In 3-D the three categories occupy distinct regions—**rocky** (stars mark class means) separates cleanly from **lakes/balanced** along PC1—so the category is present in the belief right where the door choice is made. **(b)** The same points in the PC1–PC2 plane, now with the marker showing the *door reached*: up-triangles (top) and down-triangles (bottom) segregate by category, and the rare timeouts (×) cluster together—the geometry of the belief and the geometry of the decision line up. Note the top three PCs hold only ~30% of the variance: the belief is genuinely high-dimensional, with category a modest but well-separated factor.

5.3 Experiment 2: the belief shapes the dream

We seed the world model with beliefs from mid-corridor and let it imagine forward (§4.4), then measure the terrain it hallucinates. Imagined rollouts from a **lakes** belief contain markedly more water than those from any other belief, and rollouts from a **rocky** belief contain markedly more

rock—the imagined future reflects the believed category, not the currently-visible tiles (which by mid-corridor are mostly empty grass). The belief is therefore not just a label read at decision time; it is an active generator of the content of the agent’s planning rollouts.

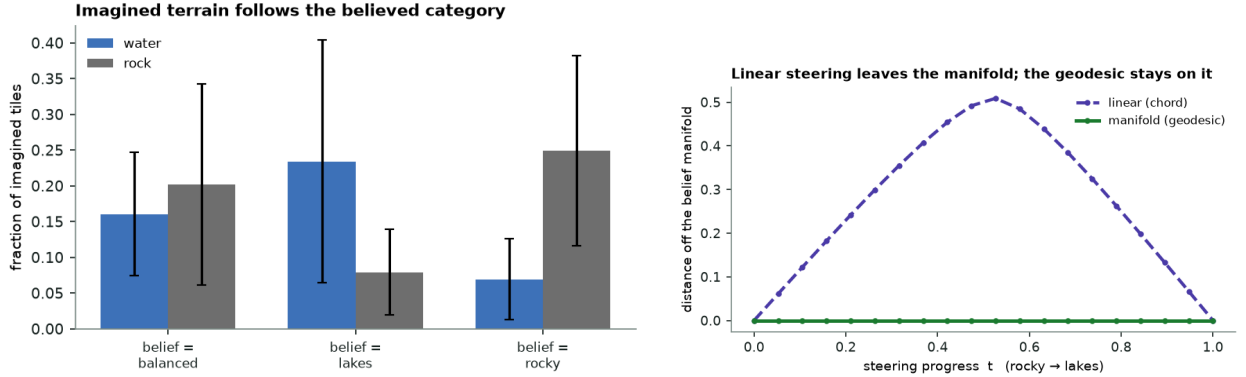


Figure 5: Left (Experiment 2): the terrain the world model *imagines* depends on the believed category—a **lakes** belief dreams more water, a **rocky** belief more rock (mean $\pm$ s.d. over imagined rollouts). Right (Experiment 3): a straight (linear) steering path bows off the fitted belief manifold at mid-traverse, while the geodesic path stays on it by construction—direct evidence that the belief is curved.

Decoded imagined future ('dream') seeded from each belief

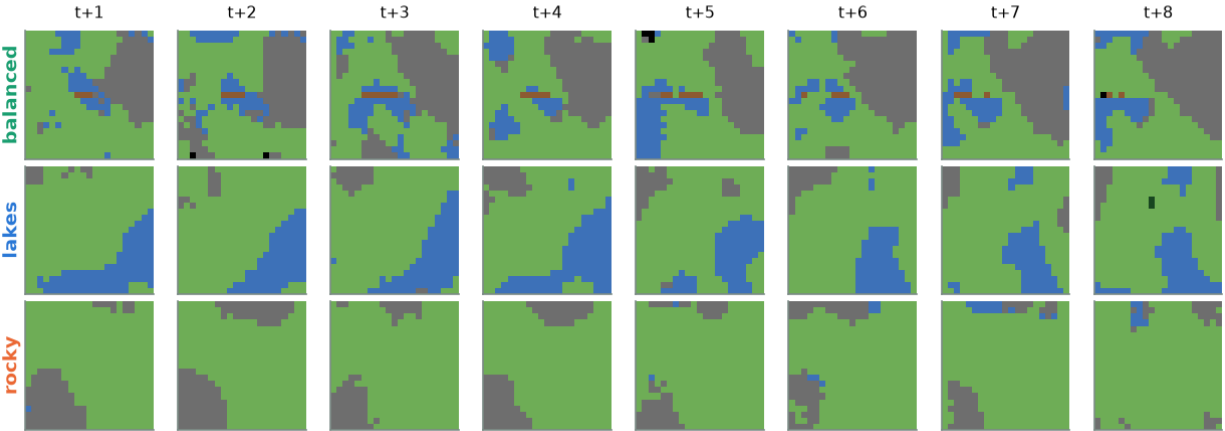


Figure 6: Experiment 2, qualitative. Decoded imagined-future (“dream”) strips, one row per belief category, seeded mid-corridor and rolled forward under the actor. The frames are legible maps whose composition matches the seeded belief.

5.4 Experiment 3: belief is a *curved* manifold, but a robust one

Using the machinery of §4.3 we ask whether belief is a direction or a manifold. The answer to the geometric question is clear: it is **curved**. A straight (linear) steering path between two decisive beliefs bows measurably off the fitted belief manifold—peaking around 0.5 in the evidence-accumulation corridor and up to ≈ 0.9 across the full door-to-door traverse—while the manifold-trace path stays at 0 off the manifold by construction. Since a straight line is exactly what direction-based steering assumes, this explains why that steering was imperfect in Experiment 1: the direction is a chord across a curved object.

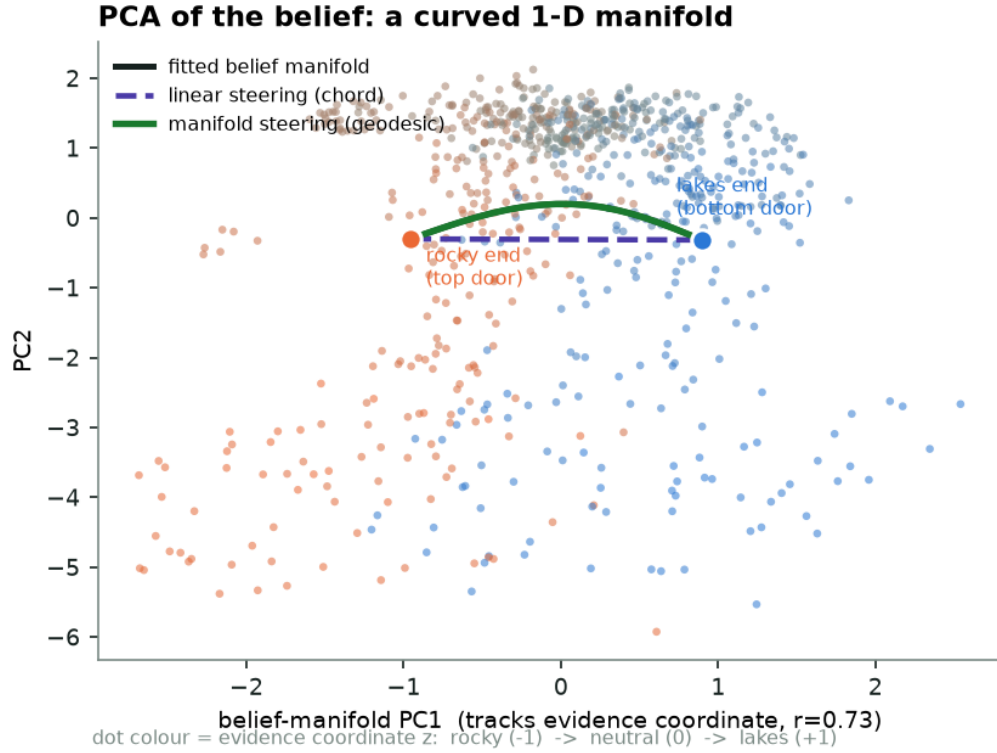


Figure 7: Experiment 3: PCA of the belief. Each dot is a real belief state, coloured by the evidence coordinate ζ (rocky $-1 \rightarrow$ neutral $0 \rightarrow$ lakes $+1$); PC1 of the manifold’s own plane tracks ζ ($r = 0.73$). The fitted spline (black) is the 1-D belief manifold. Steering from the rocky end to the lakes end along the *geodesic* (green) follows the curve, whereas the *linear* chord (dashed) cuts across it and bows off—the geometric root of why direction-based steering is imperfect.

Belief manifold in 3-D (corridor beliefs, coloured by evidence z)

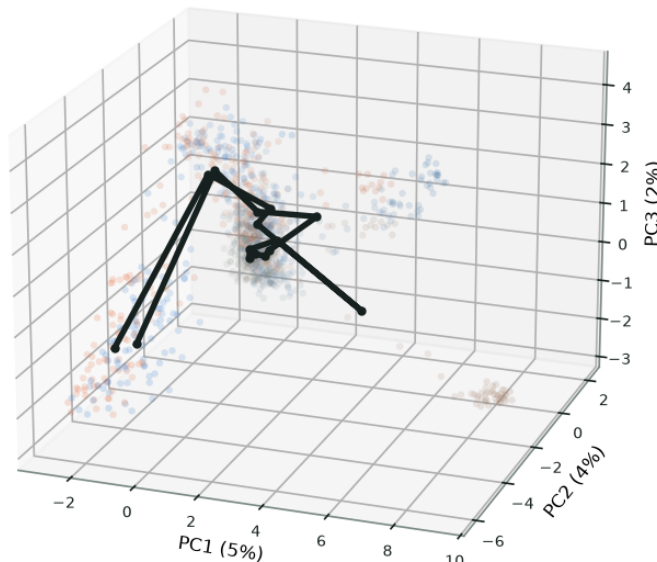


Figure 8: The same corridor belief manifold in three dimensions (a fresh 3-D PCA of the corridor belief cloud, not the planar curve basis of Fig. 7). The binned-mean polyline (black) traces a smooth curve through the cloud from the rocky pole to the lakes pole; the third component carries little additional structure, confirming the manifold is essentially one-dimensional, embedded in a high-dimensional, position-dominated space.

However, the striking downstream consequence found for a *pixel* world model—that leaving the activation manifold corrupts the decoded observation into an incoherent superposition—does *not* transfer here. On both the linear and manifold paths the decoded map stays crisp, and, when we steer the actual fork decision and imagine forward, the predicted door flips cleanly from one side to the other (the imagined commitment $p(\text{down}) - p(\text{up})$ sweeps from strongly one-sided to the other, crossing zero) with *no visible corruption*, even where the linear path sits far off the manifold. Two model-specific reasons: this RSSM belief manifold is high-dimensional and diffuse (the category coordinate is a low-variance factor inside a much larger position-dominated state, not the tidy low-dimensional closed curve a vision encoder traces), and the symbolic decoder and dynamics generalize gracefully to off-manifold beliefs rather than smearing. The honest summary is: *the belief geometry is genuinely curved, but robust*—manifold awareness confirms the shape without being required to steer it in this model.

5.5 Reading the whole picture

Putting the experiments together yields a coherent account of the agent’s mind on this task. Terrain evidence early in the corridor moves the internal belief along a curved, largely one-dimensional path parameterized by “how much the evidence favours each door”; a linear probe reads a sharpening category distribution off that path; the belief is carried, essentially unchanged, across the blind memory corridor; at the fork the actor reads the belief and commits to the matching door; and, in

the world model, the same belief generates an imagined future whose terrain and door placement match the believed category. Editing the belief edits the decision (imperfectly, because category and episode-phase are entangled), and the belief lives on a curved but robust manifold rather than a straight line.

6 Discussion and lessons

Three transferable lessons stand out.

(1) Task design controls what “memory” means. A single geometric change—confining evidence to the left of a grass corridor wider than the view radius—converts a reflex task into a working-memory task, because it forces the evidence out of the observation window before the decision. Whenever a partially-observed task is meant to test memory, one should check that the cue and the response cannot co-occur in a single observation.

(2) Discounted potential shaping can pay for loitering. Equation (2) with $\gamma_{\text{sh}} = \gamma < 1$ gives a positive per-step reward $\kappa(1 - \gamma)\text{ctg}$ for standing still, which beats a small time cost when the potential is large. Imagination-based agents fall into the resulting basin far more readily than on-policy ones. Using pure-progress shaping ($\gamma_{\text{sh}} = 1$) removes the basin at no cost to the useful gradient.

(3) Decodable, causal, and steerable are three different claims. The belief here is strongly decodable and strongly causal (behaviour tracks it almost perfectly), yet only moderately steerable by linear edits, because the representation entangles category with episode phase and curves through activation space. Interpretability claims should say which of the three they mean.

7 Conclusion

Cogniland packages a simple but pointed question—*what does an agent believe about a hidden variable, and can we read, edit, and watch it plan with that belief?*—into a small, fully-controlled navigation task and a matched set of tools. The `fork_wall` environment makes the hidden variable known, the memory requirement real, and the decision discrete; recurrent PPO and DreamerV3 give us a policy-only and a generative agent on identical data; and linear probing, activation steering, manifold geometry, and dream decoding let us turn the vague notion of “belief” into measured, falsifiable statements. The findings—belief drives the decision more tightly than ground truth, belief generates the dream, belief curves through activation space but robustly—make the environment a useful substrate for the mechanistic study of memory and planning, and the reward-shaping and task-design pitfalls we hit and fixed are lessons that generalize well beyond it.

References

- [Cho et al.(2014)] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. *EMNLP*, 2014.
- [Geiger et al.(2021)] A. Geiger, H. Lu, T. Icard, and C. Potts. Causal abstractions of neural networks. *NeurIPS*, 2021. (Framework realized in Goodfire’s `causalab`, `manifold_steering` branch.)
- [Ha & Schmidhuber(2018)] D. Ha and J. Schmidhuber. World models. *NeurIPS*, 2018.
- [Hafner et al.(2023)] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models. *arXiv:2301.04104*, 2023.

- [Kaelbling et al.(1998)] L. Kaelbling, M. Littman, and A. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 1998.
- [Lambrechts et al.(2022)] G. Lambrechts, A. Bolland, and D. Ernst. Recurrent networks, hidden states and beliefs in partially observable environments. *TMLR*, 2022.
- [Li et al.(2023)] K. Li, A. K. Hopkins, D. Bau, F. Viégas, H. Pfister, and M. Wattenberg. Emergent world representations: exploring a sequence model trained on a synthetic task. *ICLR*, 2023.
- [Mante et al.(2013)] V. Mante, D. Sussillo, K. Shenoy, and W. Newsome. Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature*, 2013.
- [Meng et al.(2022)] K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in GPT. *NeurIPS*, 2022.
- [Mikulik et al.(2020)] V. Mikulik, G. Delétang, T. McGrath, et al. Meta-trained agents implement Bayes-optimal agents. *NeurIPS*, 2020.
- [Morihiro et al.(2026)] N. Morihiro, A. Nahar, K. Bharadwaj, Y. Kato, A. Hayashi, and T. Harada. R2-Dreamer: redundancy-reduced world models without decoders or augmentation. *ICLR*, 2026.
- [Ng et al.(1999)] A. Ng, D. Harada, and S. Russell. Policy invariance under reward transformations: theory and application to reward shaping. *ICML*, 1999.
- [Ravichander et al.(2021)] A. Ravichander, Y. Belinkov, and E. Hovy. Probing the probing paradigm: does probing accuracy entail task relevance? *EACL*, 2021.
- [Reinsch(1967)] C. Reinsch. Smoothing by spline functions. *Numerische Mathematik*, 1967.
- [Schulman et al.(2017)] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv:1707.06347*, 2017.
- [Shai et al.(2024)] A. Shai, S. Marzen, P. Riechers, et al. Transformers represent belief state geometry in their residual stream. *arXiv*, 2024.
- [Sussillo & Barak(2013)] D. Sussillo and O. Barak. Opening the black box: low-dimensional dynamics in high-dimensional recurrent neural networks. *Neural Computation*, 2013.
- [Turner et al.(2023)] A. Turner, L. Thiergart, et al. Activation addition: steering language models without optimization. *arXiv*, 2023.
- [Wong & Wang(2006)] K.-F. Wong and X.-J. Wang. A recurrent network mechanism of time integration in perceptual decisions. *Journal of Neuroscience*, 2006.
- [Zou et al.(2023)] A. Zou, L. Phan, S. Chen, et al. Representation engineering: a top-down approach to AI transparency. *arXiv*, 2023.